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PIECEWISE LINEAR CLASSIFIERS WITH AN APPROPRIATE NUMBER OF HYPERPLANES

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Abstract—A new method to construct a piecewise linear classifier is proposed. This method selects an appropriate number of hyperplanes of a piecewise linear classifier by MDL (Minimum Description Length) criterion. This method constructs the hyperplanes so as to keep the local error rate for a training set under a threshold. The threshold is determined automatically by the MDL criterion so as to avoid overfitting of the classifier to the training set. This method showed results better than those of a previous method in some experiments. © 1998 Pattern Recognition Society. Published by Elsevier Science Ltd. All rights reserved.

Piecewise linear classifier	Local error rate	Minimum description length	Clustering
Prototype	Alphabetical character recognition	Japanese vowel recognition	

1. INTRODUCTION

In pattern recognition problems, nonparametric classifiers are effective when a statistical model cannot be assumed for the underlying distribution of samples. A piecewise linear classifier is a typical nonparametric classifier and approximates the true discrimination boundary by a combination of hyperplanes.

Many methods have been proposed for the construction of piecewise linear classifiers.^(1–6) Nilsson's *committee machine*⁽¹⁾ is considered to be a special form of the piecewise linear classifier. However, this method uses a complex procedure in the discrimination step and the procedure requires a large computational cost. Meisel suggested to put hyperplanes at the minimum of probability density functions.⁽²⁾ However, such an approach requires an estimation of the density function, which is difficult when the available training sample set is small. In addition, some methods using geometrical information of training samples have been proposed,^(3,4) but these methods impose severe restrictions on the shape of class regions and do not seem to be practical. Sklansky and Michelotti⁽⁵⁾ have improved these earlier methods for two-class problems. Their method is more effective and less restrictive than previous methods, and it has been extended to multi-class problems by Park and Sklansky.⁽⁶⁾

Our method is based on Park and Sklansky's method. Their method aims to separate prototypes belonging to different classes, where the prototypes are the cluster centers of the training samples of each

class. Therefore, unless the prototypes faithfully represent the training samples around them, this method will not work well. It is especially difficult to represent the training samples located at the class boundaries by prototypes. Our method, therefore, depends on the result of clustering and often fails even to discriminate the training samples properly.

There are two purposes for using a small number of prototypes instead of all training samples. The first purpose is to economize the computational cost of constructing hyperplanes. This is because the construction of optimal hyperplanes requires an optimization problem based on these prototypes to be solved. The second purpose is to accurately obtain the spatial structure of the underlying distribution of samples. All of the training samples may not be necessary for this purpose. Individual training samples, especially samples near class boundaries, are not always strong evidence of true class boundaries, since they occur randomly by nature and are often influenced by noise. Therefore, it is important to use an appropriate number of training samples simultaneously instead of individual samples. Thus, representative prototypes are effective for obtaining the global structure of samples as long as they are selected appropriately.

However, the use of prototypes is difficult for two reasons: (1) It is difficult to find an appropriate set of prototypes in a high-dimensional space, and (2) It is difficult to determine the appropriate number of prototypes. If we choose an inappropriate set of prototypes, we cannot expect a good construction of hyperplanes. The method by Park and Sklansky aims to separate all prototypes by as small a number of hyperplanes as possible. However, this sometimes produces undesirable results since it ignores local information, i.e. training samples around each

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prototype. In other words, their method sometimes chooses an insufficient number of hyperplanes.

To resolve this problem, our method uses both the prototypes and the training samples evenly and constructs hyperplanes incrementally so as to keep the error rate for the local training samples under a threshold. We can obtain a very high recognition rate in the training samples by making the value of the threshold very small. However, the high recognition rate for the training samples does not necessarily guarantee the same performance for many unknown samples. When the number of training samples is small, fitting a classifier too close to the training samples may even be harmful. Therefore, we try to find an appropriate value for the threshold on the basis of the MDL criterion.⁽⁷⁾

2. CONSTRUCTION OF PIECEWISE LINEAR CLASSIFIERS

2.1 Basic algorithm

We explain our approach along with the method proposed by Park and Sklansky.⁽⁶⁾ Both methods are carried out according to the following basic algorithm.

Step 1. Using a clustering method, e.g. Forgy's algorithm,⁽⁸⁾ we find some clusters over the training samples in each class and represent the cluster centers by *prototypes*.

Step 2. Among all the links that connect pairs of prototypes belonging to two different classes, we find the *Tomek links*,⁽⁹⁾ i.e. a link where the hypersphere with the link as the diameter does not include other prototypes. For simplicity, we refer to a Tomek link as a link.

Step 3. We find hyperplanes so as to cut all the (Tomek) links.

Step 4. We assign a class label to each region surrounded by these hyperplanes. This is done by the majority rule for the number of the training samples that fall in the region. We classify an unknown sample according to the label of the region where the sample falls in. When the region has no training samples, that is, when the region is not labeled, we reject the sample

(with-reject mode) or adopt the nearest region to the unknown sample (without-reject mode).

The essential part of this procedure is Step 3. In Step 3, Park and Sklansky's approach finds hyperplanes in such a way that each hyperplane cuts the links cut as many as possible. As a result, the method succeeds in finding a near-minimum number of hyperplanes that is sufficient for separating all the prototypes. However, there is no warranty that the hyperplanes separate all training samples as well. This problem may be solved by increasing the number of prototypes, but this causes an increase in the computational cost for finding hyperplanes, and it is difficult to find an appropriate number of prototypes.

Therefore, we take another approach. We find a (Tomek) link first and locate a hyperplane so as to correctly classify the training samples around the two ends of the link [Fig. 1(a)]. Next, we add another link [Fig. 1(b)] and train the hyperplane so as to correctly classify all the training samples around the four ends of the two links [Fig. 1(c)], as long as the error rate is kept sufficiently low. We train a hyperplane by repeating this procedure. Although our method can keep the error rate for the training samples sufficiently low, it sometimes requires more hyperplanes than Park and Sklansky's method in order to cut all the links. Their method uses a similar mechanism for training hyperplanes on the basis of training samples, but the cutting of many links has priority over training.

2.2 Incremental construction of hyperplanes

In this section we show the procedure for the construction of hyperplanes. A flow chart of the procedure is shown in Fig. 2.

Step 0. Let the set of links be L . Determine the value of the upper bound $\epsilon_{\max}$ of the local error rate.

Step 1. Repeat the following steps until L becomes empty.

Step 2. Select the longest link l in L as an initial link, and let the perpendicular bisector of the link be an initial hyperplane h . Let $L_h = \{l\}$ and $L = L - \{l\}$. Let p denote the prototype of l that is located on the

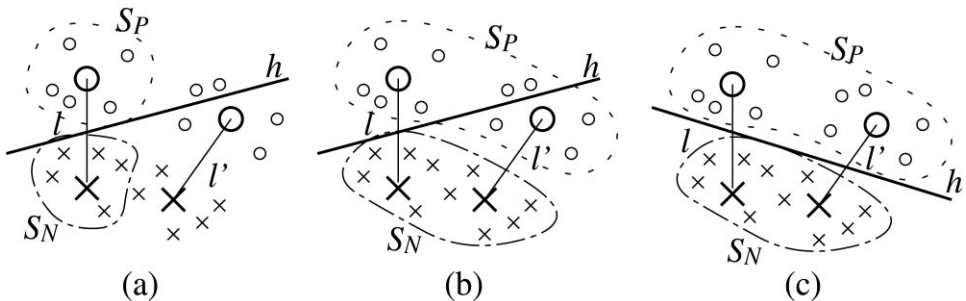


Fig. 1. Construction of hyperplanes by the proposed method. The small symbols denote the training samples and the large symbols denote the prototypes.

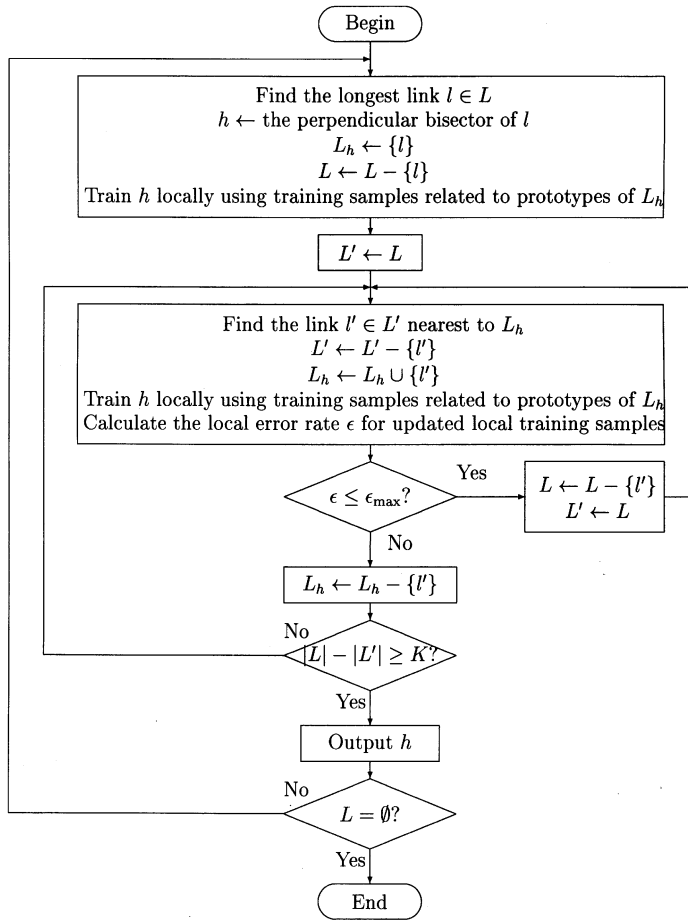


Fig. 2. Flow chart of the proposed method.

positive side with respect to h and n , the other prototype of l , that is located on the negative side. Let a *positive prototype set* $P = \{p\}$ and a *negative prototype set* $N = \{n\}$. By collecting the training samples belonging to the same cluster with $p \in P$, make a *local positive set* S_p . In a similar way, make a *local negative set* S_n . Next, train h locally so as to classify S_p and S_n more correctly using a procedure called *Window Training Procedure*.⁽¹⁰⁾ Copy L to L' .

Step 3. Find the link $l' \in L'$ nearest to L_h , where the nearness between l' and L_h is measured by distance $D(l', L_h) = \min_{l \in L_h} d(l', l)$, and $d(\cdot, \cdot)$ is the distance between centers of the two links. Next, let $L' \leftarrow L' - \{l'\}$ and $L_h \leftarrow L_h \cup \{l'\}$. If h also cuts l' simultaneously, try to add both prototypes of l' to P and N according to their signs with respect to h [Fig. 3(a)]. If both prototypes are located on the negative side with respect to h , the prototype nearer h is added to P , and the other prototype is added to N [Fig. 3(b)]. If both prototypes are located on the positive side, add them to N and P , conversely. Then, reconstruct S_p and S_n by collecting the training samples belonging to the same cluster as at least one prototype of P and N , respectively.

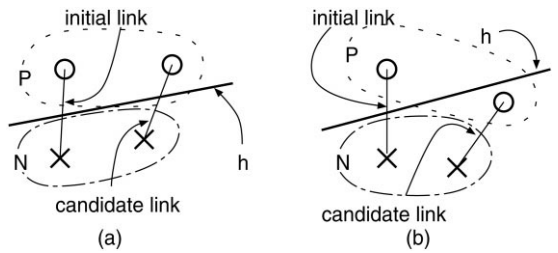


Fig. 3. Addition of a candidate link: (a) when h also cuts the candidate link, (b) when both prototypes of the candidate link are located on the negative side.

Step 4. Train h locally for S_p and S_n , and calculate the local error rate ϵ for S_p and S_n .

Step 5. If $\epsilon \leq \epsilon_{\max}$, update the two sets L and L_h as $L = L - \{l'\}$ and $L' \leftarrow L$. We call this limitation *error limitation*. Furthermore, we limit the addition of the candidate links only when at least one side of h is always a class (Fig. 4). We call this limitation *class limitation*. If both limitations are satisfied, return to Step 3. Otherwise, cancel the addition of l' as $L_h \leftarrow L_h - \{l'\}$ (P and N returned to the situation they

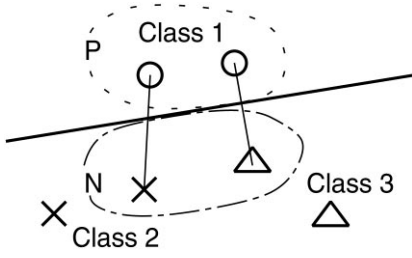


Fig. 4. Multi-class problems.

were at before the addition of l') and return to Step 3 in order to find a second nearest link to L_h . If there is no link to satisfy these limitations or if the number of canceled links is beyond a given number K , terminate the addition of links to L_h and return to Step 2 for finding another hyperplane h' .

In this algorithm, it should be noted that every initial hyperplane is adopted without the error limitation, and thus, every link will be cut sometime. This also means that these initial hyperplanes may not accurately classify the local training samples around them.

2.3. Determination of $\varepsilon_{\max}$ using the MDL criterion

In the proposed method, the recognition rate for the training samples is controllable by a threshold $\varepsilon_{\max}$, which is used as an upper bound for the local error rate. With a small value of $\varepsilon_{\max}$, we can have a classifier that can separate the training samples well. However, overfitting of a classifier to the training samples does not always improve the capability of the classifier for many unknown samples. There is an appropriate value of $\varepsilon_{\max}$ for a given problem.

To estimate an appropriate value of $\varepsilon_{\max}$, we use the MDL criterion,⁽⁷⁾ which is one of the probabilistic model selection criteria. The MDL criterion, which is based on data compression, selects a certain model M from a model class $\mathcal{M}$ such that M attains the minimum cost for describing the data, the class labels of the training samples in our model, as well as the cost for describing M itself. If we choose a perfect classifier that can classify every training sample correctly, then training samples would be explained well by the classifier but the classifier itself would be very complicated. On the other hand, if we choose a classifier that is too simple, then all of the training samples cannot be explained. In the MDL criterion, we evaluate both factors in bit-length to describe them.

The cost of describing the classifier and the classification information is denoted by

$$L_{\text{MDL}} = L(X^N | \theta) + L(\theta | M) + L(M),$$

where X^N is a given set of N training samples and θ is a real-valued parametric vector belonging to the

model M . The first term $L(X^N | \theta)$ is the description length of X^N under a particular θ of model M and is calculated as the log-likelihood of θ with respect to X^N , i.e. $-\log_2 P(X^N | \theta)$. The second term $L(\theta | M)$ is the description length of θ , and the third term $L(M)$ is the description length for model M . Summing up these terms, we have a value of L_{MDL} , and we select a model with the minimum description length. In the above formula, the first term shows how well the classifier can explain the training samples, and the second and third terms show how complicated the classifier is.

The practical calculation of these terms for our piecewise linear classifier is as follows. For the first term, the log-likelihood of θ , we consider a finite partitioning model [for example, see reference (11)]. Let us assume that the universal region $\mathcal{R}$ is partitioned into R regions $\{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_R\}$ by hyperplanes. Let N_r , N_r^+ , and N_r^- be the number of the training samples in a region $\mathcal{R}_r$, the number of the training samples of the most dominant class, and the number of training samples of other classes, respectively. Then, the maximum likelihood estimator of a binomial distribution for $\mathcal{R}_r$ is given by $\hat{\theta}_r = N_r^+ / N_r$.

Thus, the first term $L(X^N | \theta)$ is calculated by

$$\begin{aligned} & \sum_{r=1}^R -\log_2 \hat{\theta}_r^{N_r^+} (1 - \hat{\theta}_r)^{N_r^-} \\ &= \sum_{r=1}^R N_r \{ -\hat{\theta}_r \log_2 \hat{\theta}_r - (1 - \hat{\theta}_r) \log_2 (1 - \hat{\theta}_r) \} \\ &= \sum_{r=1}^R \left(-N_r^+ \log_2 \frac{N_r^+}{N_r} - N_r^- \log_2 \frac{N_r^-}{N_r} \right). \end{aligned}$$

The second term $L(\theta | M)$ is a code length for encoding θ . In this paper, we use

$$\frac{1}{2} (D + 1) H (\log_2 N + \log_2 e)$$

for the term where D is the number of features and H is the number of hyperplanes. Here, $(D + 1) H$ means the number of parameters required to describe the hyperplanes of a piecewise linear classifier. The coefficient $1/2$ means that a real value is quantized with proportion to the square root of N , and the term $\log_2 e$ is needed for the quantization error. For the last term $L(M)$, we identify a certain model M by the number of hyperplanes. Thus, a model class $\mathcal{M}$ is identical to the set of natural numbers. For this purpose, we can use the universal encoding method of integers proposed by Rissanen,⁽⁷⁾ that is, $\log_2^* H$ bits, where $\log_2^* H = 1.519 + \log_2 H + \log_2 \log_2 H + \dots$ and the summation is taken only for positive terms. By summing up these three terms, we obtain the value of L_{MDL} .

Here, we ignore the cost for describing the dominant class of a region and the cost for the relationship between hyperplanes and regions, as we regard these costs as being negligible compared to the other costs (in fact, these costs are $O(1)$ with respect to N).

2.4. Practical determination of threshold $\varepsilon_{\max}$

As mentioned in the previous section, we use the MDL criterion for determining threshold $\varepsilon_{\max}$. With increases in the number of hyperplanes, the first term decreases and the second term increases. Since L_{MDL} is the sum of the terms, it makes a trade-off curve that has one minimum point. We choose an appropriate number of hyperplanes so that L_{MDL} is minimum.

In practice, varying $\varepsilon_{\max}$ from 0 to 1 by a certain step-width, we record the number of hyperplanes M and the corresponding value of L_{MDL} , and we then find the minimum of L_{MDL} using a least square curve. In the proposed method, a change in the value of $\varepsilon_{\max}$ does not always cause a change in the number M . Therefore, when some different values of $\varepsilon_{\max}$ result in the same M , we choose the minimum value of L_{MDL} .

3. EXPERIMENTS

All experiments were carried out on an *Intel Pentium* 200 MHz machine with *BSDI BSD/OS*. For all the experiments, we determined the value of K for the terminal condition by $2D$, i.e. twice the number of features.

To determine $\varepsilon_{\max}$ by the MDL criterion, we plotted pairs of the number of hyperplanes and MDL values, varying $\varepsilon_{\max}$ from 0.02 to 0.5 by increments of 0.02. A smaller value of $\varepsilon_{\max}$ means a larger number of hyperplanes. We found a quadratic curve by applying the least square method to all the points.

For recognition, we adopted two modes: when an unknown sample falls into a region without a label, (a) we rejected it (with-reject mode) or (b) we assigned it a label by the region with a label nearest to the region (without-reject mode). We evaluated the goodness of a classifier mainly in the with-reject mode, because the with-reject mode reflects the decision boundary.

3.1. Artificial data

We carried out experiments on two-class artificial data that were distributed on double rings centered at the origin in a 2-dimensional space. The distribution of Class 1 is the normal distribution with mean

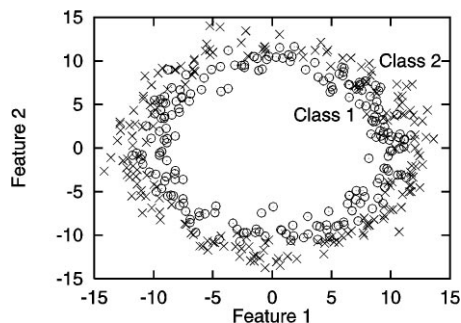


Fig. 5. Training samples for artificial data when $r_2 = 12$ and $n = 200$.

$r_1 = 10$ (fixed) and variance 1 in the radius direction, and the distribution of Class 2 is the normal distribution with mean r_2 (variable) and variance 1 in the radius direction. There is a significant overlap between the two distributions when $|r_1 - r_2|$ is small. For each class, we used n (variable) samples per class for training and 1000 samples per class for the test. The recognition rate was calculated for the test samples in the with-reject mode. Ten prototypes were used for each class. An example in the case of $r_2 = 12$ and $n = 200$ is shown in Fig. 5.

We compared the proposed method with the Park and Sklansky's method for various values of the parameters. We also examined the cases with MDL estimation and the case without MDL estimation. In the latter case, the upper bound $\varepsilon_{\max}$ was fixed at 0.1. First, we varied the value of r_2 from 10 to 17 by increments of 1 and fixed n at 200. The two distributions are almost separable in $r_2 = 17$. Next, we varied the number of training samples n with $\lfloor 10^k \rfloor$ (k is varied from 1.4 to 3.4 by increments of 0.1) and fixed r_2 at 12.

The results are shown in Fig. 6. The proposed method with MDL estimation outperformed the previous method when the two distributions were not too close to each other ($r_2 \geq 12$) and the number of training samples was not too small ($n \geq 100$). This result shows the MDL criterion works well in a preferable condition.

It is noted that the proposed method has a superior convergence property with respect to n in the

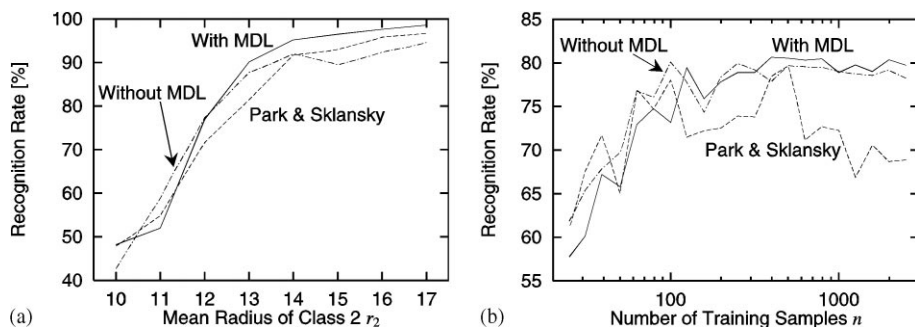


Fig. 6. Results of artificial data.

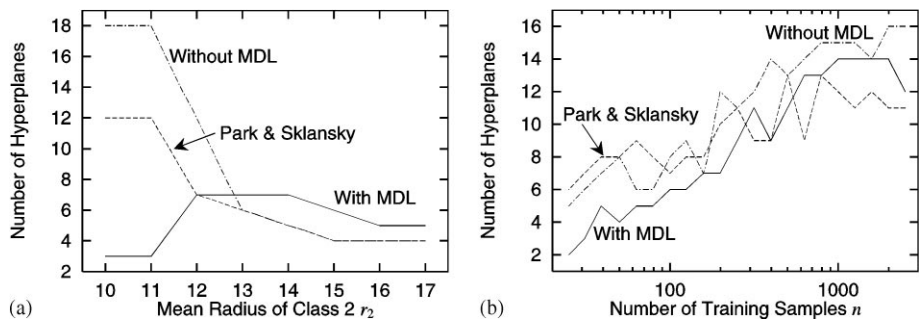


Fig. 7. Number of hyperplanes.

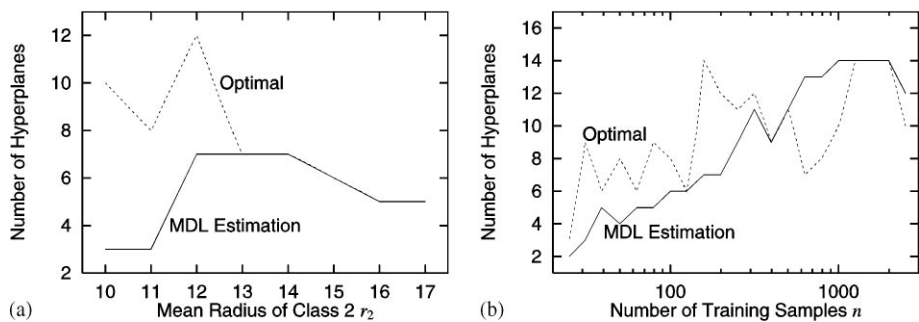


Fig. 8. Optimal number of hyperplanes.

recognition rate. In other words, the proposed method is stable for a variation in parameters.

On the other hand, when the two distributions were too close or the number of training samples was very small, the proposed method with MDL estimation did not work well.

To clarify the reason for this, we examined the numbers of hyperplanes. The results are shown in Figs 7 and 8. In these conditions, the proposed method with MDL estimation estimated a smaller number of hyperplanes than the number of hyperplanes constructed by the previous method (Fig. 7) and the number of hyperplanes constructed by the proposed method without MDL estimation. Indeed, the MDL estimation was smaller than the experimentally determined optimal number of hyperplanes (Fig. 8). In general, the MDL criterion has a tendency to underestimate the number of parameters in the case of a small number of training samples. These observations consist with the tendency.

In such cases, we recommend that a constant value (e.g. 0.1) be used for $\epsilon_{\max}$ without MDL estimation. The method in which $\epsilon_{\max}$ was fixed at 0.1 worked better than the method with MDL estimation under such conditions (Fig. 6).

3.2. Real data

We carried out experiments on (a) 26-class alphabetical character dataset taken from ETL-3 database,⁽¹²⁾ and (b) 5-class Japanese vowel data {/a/, /i/, /u/, /e/, /o/} taken from ETL-WD-I database.⁽¹⁴⁾

In Experiment (a), we used the first 10 principle components in Karhunen–Loève expansion of the original 81 features obtained using Characteristic LocI.⁽¹³⁾ For each class, we used 100 samples for training and the other 100 samples for the test. Ten prototypes were used in each class, and 693 links were found. In Experiment (b), the six features were obtained by frequency analysis. For each class, we used 100 samples for training and the other 500 samples for the test. Ten prototypes were used in each class, and 143 links were found.

The results are shown in Tables 1 and 2. In Experiment (a), the number of hyperplanes estimated by the MDL criterion is 33, which corresponds to $\epsilon_{\max}$ of 0.12. In Experiment (b), the selected number of hyperplanes is 10 and the corresponding $\epsilon_{\max}$ is 0.08.

The recognition rate of the proposed method, both in the with-reject mode and in the without-reject mode, was higher than the recognition rate of the previous method, while the number of hyperplanes in the proposed method was smaller than that in the previous method. This means the MDL criterion successfully avoided overfitting to the training samples. In addition, even when $\epsilon_{\max}$ was fixed at 0.1, the proposed method showed equivalent or better performance than that of the previous method.

Although the previous method is designed to find the nearly minimum number of hyperplanes, the proposed method chose fewer hyperplanes than the previous method. This indicates that the optimal setting of parameter values is difficult in the previous method,

Table 1. Experimental results of ETL-3 dataset

Method	M	L_{MDL}	R_1 (%)	R_2 (%)
Proposed with MDL ($\varepsilon_{\max} = 0.12$)	33	2519	75.04	93.27
Proposed without MDL (fixed $\varepsilon_{\max} = 0.1$)	35	2643	74.23	93.04
Park and Sklansky	44	3157	51.65	92.23

Here M , R_1 and R_2 are the number of hyperplanes, the recognition rate in the with-reject mode and the rate in the without-reject mode, respectively.

Table 2. Experimental results of ETL-WD-I dataset

Method	M	L_{MDL}	R_1 (%)	R_2 (%)
Proposed with MDL ($\varepsilon_{\max} = 0.08$)	10	470.7	77.88	82.44
Proposed without MDL (fixed $\varepsilon_{\max} = 0.1$)	11	505.4	77.12	81.52
Park and Sklansky	19	721.3	57.08	82.20

Here M , R_1 and R_2 are the number of hyperplanes, the recognition rate in the with-reject mode and the rate in the without-reject mode, respectively.

where these parameters are used for the optimization procedure in the method.

4. CONCLUSION

We proposed a new method to construct a piecewise linear classifier that keeps the local error rate under a threshold. The threshold is automatically determined by the MDL criterion to avoid overfitting to the training set. Through experiments, we showed the effectiveness of the proposed method.

As the case in the previous method, our method depends on the result of clustering over the training samples, i.e. the prototypes. The use of prototypes instead of every training sample is advantageous in terms of computational cost but also has the demerit that detail information of the distribution of training samples may be lost. Therefore, in a future study we will try to develop a construction method without clustering. The use of computational geometry techniques with randomized algorithms may be one possibility. Determination of the optimal number of clusters using the MDL criterion will also be discussed in a future paper.

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